SRM INSTITUTE OF SCIENCE AND TECHNOLOGY TIRUCHIRAPPALLI
TIRUCHIRAPPALLI
Department of CSE Artificial Intelligence & Machine Learning
IV Year VII Semester
Deep Learning
Challenge Based Learning (CBL)

Image Classification Challenge

50 Problem Statements with Dataset Links, Report Format and Evaluation Rubric

MODE	DURATION	DELIVERABLES	EVALUATION
Individual (one student, one problem statement)	2 hours, hands-on in the lab	Notebook with outputs + two-page report	30 marks, as per CBL rubric (Part E)

How to use this document: Part A lists all 50 problem statements. Part B gives each problem statement with its dataset link. Parts C to G — instructions to students, the two-page report format, the evaluation rubric, viva questions and notes for the lab coordinator — apply to every problem statement. Each student is allotted one problem statement by the faculty.

Part A — List of Problem Statements

PS No.	Problem Statement	Domain
PS-01	Banana Leaf Spot Disease Classification	Agriculture & Food
PS-02	Rice Leaf Disease Identification	Agriculture & Food
PS-03	Mango Leaf Disease Diagnosis	Agriculture & Food
PS-04	Sugarcane Leaf Disease Classification	Agriculture & Food
PS-05	Groundnut Leaf Disease Detection	Agriculture & Food
PS-06	Cassava (Tapioca) Leaf Disease Recognition	Agriculture & Food
PS-07	Bean Leaf Lesion Classification	Agriculture & Food
PS-08	Weed Species Recognition for Precision Spraying	Agriculture & Food
PS-09	Plant Seedling Species Classification	Agriculture & Food
PS-10	Indian Medicinal Plant Identification from Leaves	Agriculture & Food
PS-11	Rice Grain Variety Classification	Agriculture & Food
PS-12	Fresh vs Rotten Fruit Sorting	Agriculture & Food
PS-13	Pneumonia Detection from Paediatric Chest X-rays	Healthcare & Medical Imaging
PS-14	Brain Tumour Classification from MRI	Healthcare & Medical Imaging
PS-15	Skin Lesion Classification (HAM10000)	Healthcare & Medical Imaging

PS No.	Problem Statement	Domain
PS-16	Retinal OCT Disease Classification	Healthcare & Medical Imaging
PS-17	Malaria Parasite Detection in Blood Smear Cells	Healthcare & Medical Imaging
PS-18	Breast Cancer Histopathology Classification	Healthcare & Medical Imaging
PS-19	Peripheral Blood Cell Type Recognition	Healthcare & Medical Imaging
PS-20	Lung and Colon Cancer Histopathology (LC25000)	Healthcare & Medical Imaging
PS-21	Kidney Abnormality Detection from CT	Healthcare & Medical Imaging
PS-22	Knee Osteoarthritis Severity Grading	Healthcare & Medical Imaging
PS-23	COVID-19 and Lung Disease Detection from Chest X-rays	Healthcare & Medical Imaging
PS-24	Mpox Skin Lesion Screening	Healthcare & Medical Imaging
PS-25	Diabetic Retinopathy Grading from Fundus Images	Healthcare & Medical Imaging
PS-26	Land Use and Land Cover Classification from Sentinel-2	Environment & Remote Sensing
PS-27	Aerial Scene Classification	Environment & Remote Sensing
PS-28	Ship Detection in Satellite Image Chips	Environment & Remote Sensing
PS-29	Weather Condition Recognition from Outdoor Images	Environment & Remote Sensing
PS-30	Sky and Cloud Type Classification	Environment & Remote Sensing
PS-31	Waste Segregation at Source	Environment & Remote Sensing
PS-32	Casting Defect Inspection	Industry & Infrastructure
PS-33	Concrete Surface Crack Detection	Industry & Infrastructure
PS-34	Steel Strip Surface Defect Classification	Industry & Infrastructure
PS-35	Pet Breed Identification	Wildlife & Nature
PS-36	Dog Breed Recognition	Wildlife & Nature
PS-37	Bird Species Recognition	Wildlife & Nature

PS No.	Problem Statement	Domain
PS-38	Flower Species Recognition	Wildlife & Nature
PS-39	Handwritten Tamil Character Recognition	Indian Scripts, Signs & Currency
PS-40	Handwritten Devanagari Character Recognition	Indian Scripts, Signs & Currency
PS-41	Kannada Handwritten Digit Recognition	Indian Scripts, Signs & Currency
PS-42	Indian Sign Language (ISL) Recognition	Indian Scripts, Signs & Currency
PS-43	American Sign Language Alphabet Recognition	Indian Scripts, Signs & Currency
PS-44	Traffic Sign Recognition	Indian Scripts, Signs & Currency
PS-45	Indian Currency Note Recognition for the Visually Impaired	Indian Scripts, Signs & Currency
PS-46	Indian Coin Denomination Recognition	Indian Scripts, Signs & Currency
PS-47	Facial Expression Recognition	Human-Centred Vision
PS-48	Distracted Driver Detection	Human-Centred Vision
PS-49	Hand Gesture Recognition for Touchless Control	Human-Centred Vision
PS-50	Natural Scene Classification	Scene Understanding

Part B — Problem Statements

Every problem statement is an image classification task to be completed individually in the lab. The dataset link opens the official dataset page; the faculty will provide a local copy.

AGRICULTURE & FOOD (PS-01 to PS-12)

PS-01 Banana Leaf Spot Disease Classification

Domain	Agriculture & Food
Problem Statement	Banana growers lose yield when leaf spot diseases are not caught early. Build a CNN-based classifier that identifies the disease from a smartphone photo of a banana leaf. Use only the Original image set: the Augmented set is generated from the same photos and will leak into your test split if mixed in.
Classes	4 — Healthy, Sigatoka, Cordana, Pestalotiopsis
Dataset	BananaLSD (Mendeley Data) — 937 original field images, 224 × 224
Dataset Link	https://data.mendeley.com/datasets/9tb7k297ff/1

PS-02 Rice Leaf Disease Identification

Domain	Agriculture & Food
Problem Statement	Paddy is the main crop of the Cauvery delta, and leaf diseases spread quickly once they appear. Develop a deep learning model that identifies the disease from a rice leaf image so that farmers can act early. Some diseases look alike (e.g., blast and brown spot), so report per-class recall and explain the main confusions.
Classes	4 — Bacterial blight, Blast, Brown spot, Tungro
Dataset	Rice Leaf Disease Image Samples (Mendeley Data) — 5,932 images
Dataset Link	https://data.mendeley.com/datasets/fwcj7stb8r

PS-03 Mango Leaf Disease Diagnosis

Domain	Agriculture & Food
Problem Statement	Build an image classifier that separates healthy mango leaves from seven common diseases. Only about 1,800 of the 4,000 images are distinct leaves; the rest are zoomed or rotated copies. Discuss how near-duplicate images falling into both the training and test sets could inflate your accuracy.
Classes	8 — Anthracnose, Bacterial canker, Cutting weevil, Die back, Gall midge, Powdery mildew, Sooty mould, Healthy
Dataset	MangoLeafBD (Mendeley Data) — 4,000 images, 240 × 320
Dataset Link	https://data.mendeley.com/datasets/hxsnvwty3r/1

PS-04 Sugarcane Leaf Disease Classification

Domain	Agriculture & Food
Problem Statement	Sugarcane is a major cash crop in Tamil Nadu. Train a CNN that classifies field images of sugarcane leaves into disease, dried and healthy categories. The images are large (1000 × 1000), so justify the input resolution you choose and its effect on accuracy and training time.
Classes	7 — five disease categories, dried leaves, healthy leaves (as per the dataset folders)
Dataset	Sugarcane Disease Dataset (Mendeley Data) — 3,358 images
Dataset Link	https://data.mendeley.com/datasets/7fbnxcbnhp/2

PS-05 Groundnut Leaf Disease Detection

Domain	Agriculture & Food
Problem Statement	Groundnut is widely grown in the rain-fed areas of South India. Build a classifier for groundnut leaf images collected from farm fields in Karnataka. The classes are unbalanced (rust and nutrition deficiency have far fewer images), so use and report a metric that is fair to the minority classes.
Classes	5 — Early leaf spot, Late leaf spot, Rust, Nutrition deficiency, Healthy
Dataset	Dataset of groundnut plant leaf images (Mendeley Data) — about 3,000 original images
Dataset Link	https://data.mendeley.com/datasets/22p2vcbxfk/3

PS-06 Cassava (Tapioca) Leaf Disease Recognition

Domain	Agriculture & Food
Problem Statement	Cassava is an important tuber crop in Tamil Nadu. Train a CNN that recognises four cassava diseases and healthy leaves from field photographs taken by farmers, where backgrounds, lighting and leaf position vary widely. Explain which disease is hardest to detect and why.
Classes	5 — Bacterial blight, Brown streak disease, Green mottle, Mosaic disease, Healthy
Dataset	Cassava (iCassava 2019) via TensorFlow Datasets — about 9,400 labelled images
Dataset Link	https://www.tensorflow.org/datasets/catalog/cassava

PS-07 Bean Leaf Lesion Classification

Domain	Agriculture & Food
Problem Statement	With only about 1,300 images, this is a small-data problem. Build a classifier that tells healthy bean leaves from angular leaf spot and bean rust, and show how you prevent overfitting when so few training images are available.
Classes	3 — Angular leaf spot, Bean rust, Healthy
Dataset	Beans (Makerere AI Lab) via TensorFlow Datasets — 1,295 images
Dataset Link	https://www.tensorflow.org/datasets/catalog/beans

PS-08 Weed Species Recognition for Precision Spraying

Domain	Agriculture & Food
Problem Statement	Targeted herbicide spraying needs a camera that can tell a weed from its surroundings. Classify field images into eight weed species or a 'negative' (no target weed) class. About half of the images are negatives, so accuracy alone can mislead; report per-class results.
Classes	9 — eight weed species + Negative
Dataset	DeepWeeds — 17,509 images
Dataset Link	https://github.com/AlexOlsen/DeepWeeds

PS-09 Plant Seedling Species Classification

Domain	Agriculture & Food
Problem Statement	Telling crop seedlings from weed seedlings at an early stage lets farmers remove weeds before they compete with the crop. Classify top-view seedling images into 12 species, several of which look very similar at this growth stage (a fine-grained problem).
Classes	12 — crop and weed species (e.g., Maize, Sugar beet, Common chickweed, Black-grass)
Dataset	Plant Seedlings Classification (Kaggle competition, Aarhus University) — 4,750 training images
Dataset Link	https://www.kaggle.com/c/plant-seedlings-classification

PS-10 Indian Medicinal Plant Identification from Leaves

Domain	Agriculture & Food
Problem Statement	Many Ayurveda and Siddha herbs look alike, and misidentification can be harmful. Build a CNN that identifies the medicinal plant species from a single leaf image. With only about 50 images per species, explain how you get a reliable model from so little data.
Classes	30 medicinal plant species (e.g., Sandalwood, Indian mint, Oriental mustard)
Dataset	Medicinal Leaf Dataset (Mendeley Data) — about 1,500 images
Dataset Link	https://data.mendeley.com/datasets/nnytj2v3n5/1

PS-11 Rice Grain Variety Classification

Domain	Agriculture & Food
Problem Statement	Rice mills and exporters need to check the purity of a grain lot. Classify single-grain images into five rice varieties. The full dataset is large (75,000 images), so work on a balanced subset and state how many images per class you used.
Classes	5 — Arborio, Basmati, Ipsala, Jasmine, Karacadag
Dataset	Rice Image Dataset (Kaggle) — 75,000 images, 15,000 per variety
Dataset Link	https://www.kaggle.com/datasets/muratkokludataset/rice-image-dataset

PS-12 Fresh vs Rotten Fruit Sorting

Domain	Agriculture & Food
Problem Statement	Automated sorting lines must reject spoiled fruit before packing. Build a classifier for apples, bananas and oranges in fresh and rotten condition, and check whether your model has learnt spoilage cues or is simply recognising the fruit type.
Classes	6 — Fresh / Rotten × Apple, Banana, Orange
Dataset	Fruits Fresh and Rotten for Classification (Kaggle)
Dataset Link	https://www.kaggle.com/datasets/sriramr/fruits-fresh-and-rotten-for-classification

HEALTHCARE & MEDICAL IMAGING (PS-13 to PS-25)

PS-13 Pneumonia Detection from Paediatric Chest X-rays

Domain	Healthcare & Medical Imaging
Problem Statement	Classify chest X-rays of children as Normal or Pneumonia. The supplied validation folder has only 16 images and the classes are imbalanced, so build your own validation split from the training data and report sensitivity (recall for pneumonia) along with accuracy.
Classes	2 — Normal, Pneumonia
Dataset	Chest X-Ray Images (Pneumonia), Kaggle — 5,863 images
Dataset Link	https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

PS-14 Brain Tumour Classification from MRI

Domain	Healthcare & Medical Imaging
Problem Statement	Classify brain MRI slices into three tumour types or no tumour. The dataset combines images from three public sources, so study your misclassified images and discuss whether the model may be picking up source-specific cues rather than tumour features.
Classes	4 — Glioma, Meningioma, Pituitary tumour, No tumour
Dataset	Brain Tumor MRI Dataset (Kaggle) — about 7,000 images
Dataset Link	https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

PS-15 Skin Lesion Classification (HAM10000)

Domain	Healthcare & Medical Imaging
Problem Statement	Classify dermatoscopic images into seven types of pigmented skin lesion. Melanocytic nevi form about two-thirds of the data and some lesions have more than one image, so split by lesion ID (given in the metadata) and report macro F1 rather than accuracy.
Classes	7 — Actinic keratosis, Basal cell carcinoma, Benign keratosis, Dermatofibroma, Melanoma, Melanocytic nevus, Vascular lesion
Dataset	HAM10000 (Harvard Dataverse) — 10,015 images
Dataset Link	https://doi.org/10.7910/DVN/DBW86T

PS-16 Retinal OCT Disease Classification

Domain	Healthcare & Medical Imaging
Problem Statement	Optical coherence tomography (OCT) scans are used to diagnose sight-threatening retinal disease. Classify OCT scans into three disease classes and normal. Work on a balanced subset (the full set has over 80,000 images) and keep all scans of one patient in the same split; the patient ID is part of each file name.
Classes	4 — CNV, DME, Drusen, Normal
Dataset	Labeled OCT and Chest X-Ray Images (Mendeley Data, Kermany et al.) — OCT folder
Dataset Link	https://data.mendeley.com/datasets/rscbjbr9sj/3

PS-17 Malaria Parasite Detection in Blood Smear Cells

Domain	Healthcare & Medical Imaging
Problem Statement	Microscopists in endemic areas screen thousands of cells every day. Build a CNN that classifies segmented red blood cell images as parasitized or uninfected, and aim for a small model that could run on a low-cost device attached to a microscope.
Classes	2 — Parasitized, Uninfected
Dataset	Malaria cell images (NIH) via TensorFlow Datasets — 27,558 images
Dataset Link	https://www.tensorflow.org/datasets/catalog/malaria

PS-18 Breast Cancer Histopathology Classification

Domain	Healthcare & Medical Imaging
Problem Statement	Classify microscopic images of breast tumour tissue as benign or malignant. The images come from 82 patients at four magnifications (40×, 100×, 200×, 400×); analyse whether accuracy changes with magnification and discuss the effect of the same patient appearing in both training and test sets.
Classes	2 — Benign, Malignant (eight sub-types also available)
Dataset	BreakHis, resized 224 × 224 version (Mendeley Data) — 7,909 images
Dataset Link	https://data.mendeley.com/datasets/jxwvdwhpc2/1

PS-19 Peripheral Blood Cell Type Recognition

Domain	Healthcare & Medical Imaging
Problem Statement	Automated differential counts speed up haematology laboratories. Classify microscope images of individual normal blood cells into eight types. The immature granulocyte class groups three sub-types together, which makes it harder; check its recall and explain the errors.
Classes	8 — Neutrophil, Eosinophil, Basophil, Lymphocyte, Monocyte, Immature granulocyte, Erythroblast, Platelet
Dataset	Microscopic peripheral blood cell images (Mendeley Data, Hospital Clinic of Barcelona) — 17,092 images
Dataset Link	https://data.mendeley.com/datasets/snkd93bnjr/1

PS-20 Lung and Colon Cancer Histopathology (LC25000)

Domain	Healthcare & Medical Imaging
Problem Statement	Classify histopathology images into five lung and colon tissue classes. All 25,000 images were generated by augmenting 1,250 original images, so near-identical images will appear in both training and test sets; explain how this could inflate your reported accuracy.
Classes	5 — Lung adenocarcinoma, Lung squamous cell carcinoma, Benign lung tissue, Colon adenocarcinoma, Benign colon tissue
Dataset	LC25000 — 25,000 images
Dataset Link	https://github.com/tampapath/lung_colon_image_set

PS-21 Kidney Abnormality Detection from CT

Domain	Healthcare & Medical Imaging
Problem Statement	Classify kidney CT slices as normal or showing a cyst, stone or tumour. Many slices come from the same patient study, so a random split can leak information; discuss how this affects your accuracy. Stone is the smallest class, so report its recall separately.
Classes	4 — Normal, Cyst, Stone, Tumour
Dataset	CT Kidney Dataset (Kaggle) — 12,446 images
Dataset Link	https://www.kaggle.com/datasets/nazmul0087/ct-kidney-dataset-normal-cyst-tumor-and-stone

PS-22 Knee Osteoarthritis Severity Grading

Domain	Healthcare & Medical Imaging
Problem Statement	Grade knee X-rays on the Kellgren–Lawrence (KL) scale from 0 (healthy) to 4 (severe). Neighbouring grades are hard to tell apart even for radiologists, so along with accuracy report how many of your errors are off by only one grade.
Classes	5 — KL grade 0, 1, 2, 3, 4
Dataset	Knee Osteoarthritis Severity Grading Dataset (Mendeley Data, from the OAI cohort)
Dataset Link	https://data.mendeley.com/datasets/56rmx5bjcr

PS-23 COVID-19 and Lung Disease Detection from Chest X-rays

Domain	Healthcare & Medical Imaging
Problem Statement	Classify chest X-rays into COVID-19, lung opacity, viral pneumonia and normal. The images were pooled from many hospitals and sources, so use a saliency method such as Grad-CAM to check that the model looks at the lungs and not at text markers or image borders (shortcut learning).
Classes	4 — COVID-19, Lung opacity, Viral pneumonia, Normal
Dataset	COVID-19 Radiography Database (Kaggle) — about 21,000 images
Dataset Link	https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database

PS-24 Mpox Skin Lesion Screening

Domain	Healthcare & Medical Imaging
Problem Statement	Mpox lesions can look like chickenpox or measles. Build a binary classifier (Mpox vs Others) from only 228 original images. Never let an augmented copy of a test image appear in training, and justify every step you take against overfitting.
Classes	2 — Monkeypox, Others (chickenpox, measles)
Dataset	Monkeypox Skin Lesion Dataset (MSLD), Kaggle — 228 original images
Dataset Link	https://www.kaggle.com/datasets/nafin59/monkeypox-skin-lesion-dataset

PS-25 Diabetic Retinopathy Grading from Fundus Images

Domain	Healthcare & Medical Imaging
Problem Statement	Diabetic retinopathy is a leading cause of preventable blindness in India. Grade retinal fundus photographs (collected at Aravind Eye Hospital) into five severity levels. The grades are ordered and imbalanced, so report quadratic weighted kappa along with accuracy.
Classes	5 — No DR, Mild, Moderate, Severe, Proliferative DR
Dataset	APTOS 2019 Blindness Detection (Kaggle competition) — 3,662 labelled images
Dataset Link	https://www.kaggle.com/c/aptos2019-blindness-detection

ENVIRONMENT & REMOTE SENSING (PS-26 to PS-31)

PS-26 Land Use and Land Cover Classification from Sentinel-2

Domain	Environment & Remote Sensing
Problem Statement	Classify 64 × 64 Sentinel-2 satellite patches into ten land-use classes for town planning and crop monitoring. Work with the RGB version and identify the class pairs your model confuses most (e.g., highway vs river, pasture vs herbaceous vegetation).
Classes	10 — Annual crop, Forest, Herbaceous vegetation, Highway, Industrial, Pasture, Permanent crop, Residential, River, Sea/Lake
Dataset	EuroSAT — 27,000 labelled patches
Dataset Link	https://github.com/phelber/EuroSAT

PS-27 Aerial Scene Classification

Domain	Environment & Remote Sensing
Problem Statement	Classify high-resolution aerial images into 21 land-use scenes. With only 100 images per class, discuss how you obtained good accuracy from limited data, and analyse scenes that share content (e.g., dense vs medium residential).
Classes	21 — e.g., Agricultural, Airplane, Beach, Buildings, Dense residential, Freeway, Harbour, Parking lot
Dataset	UC Merced Land Use Dataset — 2,100 images, 256 × 256
Dataset Link	http://weegee.vision.ucmerced.edu/datasets/landuse.html

PS-28 Ship Detection in Satellite Image Chips

Domain	Environment & Remote Sensing
Problem Statement	Monitoring ports and coastal waters requires spotting ships in satellite imagery. Classify 80 × 80 image chips as ship or no-ship. The data is imbalanced (1 : 3) and some no-ship chips contain partial ships, so report precision and recall for the ship class.
Classes	2 — Ship, No-ship
Dataset	Ships in Satellite Imagery (Kaggle, Planet imagery) — 4,000 chips
Dataset Link	https://www.kaggle.com/datasets/rhammell/ships-in-satellite-imagery

PS-29 Weather Condition Recognition from Outdoor Images

Domain	Environment & Remote Sensing
Problem Statement	Traffic and surveillance cameras can adapt their processing if they know the current weather. Classify outdoor images into four weather conditions using about 1,100 images, and discuss where the 'sunrise' and 'shine' classes overlap.
Classes	4 — Cloudy, Rain, Shine, Sunrise
Dataset	Multi-class Weather Dataset (Mendeley Data) — 1,125 images
Dataset Link	https://data.mendeley.com/datasets/4drtyfjtfy/1

PS-30 Sky and Cloud Type Classification

Domain	Environment & Remote Sensing
Problem Statement	Ground-based sky cameras support solar-power forecasting and weather observation. Classify sky image patches into six cloud categories, and discuss whether texture or colour carries more of the information your model uses.
Classes	6 — Clear sky, Patterned clouds, Thick dark clouds, Thin white clouds, Thick white clouds, Veil clouds
Dataset	SWIMCAT-ext (Mendeley Data) — 2,100 images
Dataset Link	https://data.mendeley.com/datasets/vwdd9grvdp/1

PS-31 Waste Segregation at Source

Domain	Environment & Remote Sensing
Problem Statement	Smart bins and material recovery facilities need to sort dry waste automatically. Classify single waste items into six material classes. The 'trash' class is much smaller than the others, so report per-class results and explain its errors.
Classes	6 — Glass, Paper, Cardboard, Plastic, Metal, Trash
Dataset	TrashNet — 2,527 images
Dataset Link	https://github.com/garythung/trashnet

INDUSTRY & INFRASTRUCTURE (PS-32 to PS-34)

PS-32 Casting Defect Inspection

Domain	Industry & Infrastructure
Problem Statement	Replace manual visual inspection of submersible pump impeller castings. Classify top-view images as defective or OK. A missed defect costs far more than a false alarm, so choose a decision threshold that reflects this and report the resulting recall for the defective class.
Classes	2 — Defective, OK
Dataset	Casting product image data for quality inspection (Kaggle) — about 7,300 grayscale images, 300 × 300
Dataset Link	https://www.kaggle.com/datasets/ravirajsinh45/real-life-industrial-dataset-of-casting-product

PS-33 Concrete Surface Crack Detection

Domain	Industry & Infrastructure
Problem Statement	Inspecting buildings and bridges for cracks is slow and risky. Classify concrete surface patches as cracked or not cracked. The task is easy for a large network, so aim for the smallest, fastest model that still reaches high accuracy, and report its size and inference time.
Classes	2 — Crack (positive), No crack (negative)
Dataset	Concrete Crack Images for Classification (Mendeley Data) — 40,000 images, 227 × 227
Dataset Link	https://data.mendeley.com/datasets/5y9wdsg2zt/2

PS-34 Steel Strip Surface Defect Classification

Domain	Industry & Infrastructure
Problem Statement	Classify grayscale images of hot-rolled steel strip into six defect types for automated quality control. Only 300 images per class are available; some defects look very different within a class while others (rolled-in scale, crazing, pitted surface) look alike across classes.
Classes	6 — Rolled-in scale, Patches, Crazing, Pitted surface, Inclusion, Scratches
Dataset	NEU Surface Defect Database, NEU-CLS (Northeastern University) — 1,800 images
Dataset Link	https://faculty.neu.edu.cn/songkc/en/zdylm/263265/list/index.htm

WILDLIFE & NATURE (PS-35 to PS-38)

PS-35 Pet Breed Identification

Domain	Wildlife & Nature
Problem Statement	Classify photographs of cats and dogs into 37 breeds. With about 200 images per breed and breeds that differ only in fine details, this is a fine-grained problem; identify the breed pairs your model confuses and explain why.
Classes	37 — 12 cat breeds and 25 dog breeds
Dataset	Oxford-IIIT Pet Dataset — about 7,400 images
Dataset Link	https://www.robots.ox.ac.uk/~vgg/data/pets/

PS-36 Dog Breed Recognition

Domain	Wildlife & Nature
Problem Statement	Classify dog images into 120 breeds. This dataset was built from ImageNet, so an ImageNet-pretrained network may already have seen some of your test images; discuss what this means for the accuracy you report.
Classes	120 dog breeds
Dataset	Stanford Dogs Dataset — 20,580 images
Dataset Link	http://vision.stanford.edu/aditya86/ImageNetDogs/

PS-37 Bird Species Recognition

Domain	Wildlife & Nature
Problem Statement	Fine-grained bird recognition supports biodiversity surveys. Classify 200 bird species with only about 60 images per species. Use the bounding boxes provided with the dataset to study whether cropping to the bird improves accuracy.
Classes	200 bird species (mostly North American)
Dataset	Caltech-UCSD Birds-200-2011 (CUB-200-2011) — 11,788 images
Dataset Link	https://www.vision.caltech.edu/datasets/cub_200_2011/

PS-38 Flower Species Recognition

Domain	Wildlife & Nature
Problem Statement	Classify flower photographs into 102 categories. Flowers vary widely in pose, scale and lighting, and the official split gives only 10 training images per class; use the official split and explain how you cope with so few training images.
Classes	102 flower categories
Dataset	Oxford 102 Category Flower Dataset — 8,189 images
Dataset Link	https://www.robots.ox.ac.uk/~vgg/data/flowers/102/

INDIAN SCRIPTS, SIGNS & CURRENCY (PS-39 to PS-46)

PS-39 Handwritten Tamil Character Recognition

Domain	Indian Scripts, Signs & Currency
Problem Statement	Tamil OCR for forms and old records needs reliable character recognition. Classify isolated handwritten Tamil characters into 156 classes using the offline (bi-level TIFF) version of the dataset, and discuss which characters get confused and why (e.g., characters that differ by one small stroke).
Classes	156 Tamil characters
Dataset	HP Labs India Isolated Handwritten Tamil Character Dataset (hpl-tamil-iso-char) — about 500 samples per character
Dataset Link	https://lipitk.sourceforge.net/datasets/tamilchardata.htm

PS-40 Handwritten Devanagari Character Recognition

Domain	Indian Scripts, Signs & Currency
Problem Statement	Classify 32 × 32 grayscale images of handwritten Devanagari characters into 36 consonants and 10 digits. The images are small and plentiful, so design a compact CNN from scratch and justify each architectural choice.
Classes	46 — 36 consonants, 10 numerals
Dataset	Devanagari Handwritten Character Dataset (UCI) — 92,000 images
Dataset Link	https://archive.ics.uci.edu/dataset/389/devanagari+handwritten+character+dataset

PS-41 Kannada Handwritten Digit Recognition

Domain	Indian Scripts, Signs & Currency
Problem Statement	Classify handwritten Kannada digits (0–9). Besides the standard test set, evaluate on the Dig-MNIST set written by different volunteers and explain the drop in accuracy in terms of domain shift.
Classes	10 — Kannada digits 0–9
Dataset	Kannada-MNIST (Kaggle competition) — 60,000 training images plus the Dig-MNIST set
Dataset Link	https://www.kaggle.com/c/Kannada-MNIST

PS-42 Indian Sign Language (ISL) Recognition

Domain	Indian Scripts, Signs & Currency
Problem Statement	Help bridge the communication gap for the hearing- and speech-impaired by recognising static Indian Sign Language signs for English alphabets and numbers. The dataset is small and has both one- and two-handed signs; justify your augmentation choices (e.g., whether a horizontal flip changes a sign).
Classes	Static ISL signs for English alphabets and numbers (motion signs J and Z excluded)
Dataset	ISLAN (Mendeley Data) — 700 sign images
Dataset Link	https://data.mendeley.com/datasets/rc349j45m5/1

PS-43 American Sign Language Alphabet Recognition

Domain	Indian Scripts, Signs & Currency
Problem Statement	Classify hand-sign images into 26 letters plus 'space', 'delete' and 'nothing'. The images were captured against similar backgrounds, so also test your model on a few photos of your own hand taken in the lab and report how well it generalises.
Classes	29 — A–Z, Space, Delete, Nothing
Dataset	ASL Alphabet (Kaggle) — 87,000 images, 200 × 200
Dataset Link	https://www.kaggle.com/datasets/grassknoted/asl-alphabet

PS-44 Traffic Sign Recognition

Domain	Indian Scripts, Signs & Currency
Problem Statement	Driver-assistance systems must read traffic signs in poor light, with motion blur and at low resolution. Classify cropped traffic-sign images into 43 classes, and examine the errors on the rarest classes and on similar-looking signs such as speed limits.
Classes	43 traffic-sign classes
Dataset	German Traffic Sign Recognition Benchmark (GTSRB) — over 50,000 images
Dataset Link	https://benchmark.ini.rub.de/gtsrb_dataset.html

PS-45 Indian Currency Note Recognition for the Visually Impaired

Domain	Indian Scripts, Signs & Currency
Problem Statement	Build a model that recognises the denomination of an Indian currency note from a phone photo, as the core of an assistive app for visually impaired users. Notes appear in old and new designs, at different angles and backgrounds; any confusion between denominations has a direct money cost, so analyse the confusion matrix carefully.
Classes	7 denominations — ₹10, ₹20, ₹50, ₹100, ₹200, ₹500, ₹2000
Dataset	Indian Currency Dataset (Mendeley Data) — 1,786 images
Dataset Link	https://data.mendeley.com/datasets/8ckhkssyn3

PS-46 Indian Coin Denomination Recognition

Domain	Indian Scripts, Signs & Currency
Problem Statement	Coins of the same value come in several designs, while coins of different values can look alike. Classify phone images of Indian coins into four denominations and discuss which design variants cause the most errors.
Classes	4 — ₹1, ₹2, ₹5, ₹10
Dataset	Indian Coin Currency Dataset (Mendeley Data) — 1,907 images, 256 × 256
Dataset Link	https://data.mendeley.com/datasets/k49x47v732/1

HUMAN-CENTRED VISION (PS-47 to PS-49)

PS-47 Facial Expression Recognition

Domain	Human-Centred Vision
Problem Statement	Classify 48 × 48 grayscale face images into seven emotions. Human agreement on this dataset is itself limited and 'disgust' has very few images, so report per-class recall and discuss the ethical limits of automatic emotion recognition.
Classes	7 — Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral
Dataset	FER-2013 (Kaggle) — 35,887 images
Dataset Link	https://www.kaggle.com/datasets/msambare/fer2013

PS-48 Distracted Driver Detection

Domain	Human-Centred Vision
Problem Statement	Classify in-car camera images of drivers into ten behaviours such as safe driving, texting, talking on the phone or reaching behind. Use the provided driver list to keep each driver in only one split; a random split makes the task look much easier than it is.
Classes	10 — c0 (safe driving) to c9 (talking to passenger)
Dataset	State Farm Distracted Driver Detection (Kaggle competition) — 22,424 labelled images
Dataset Link	https://www.kaggle.com/c/state-farm-distracted-driver-detection

PS-49 Hand Gesture Recognition for Touchless Control

Domain	Human-Centred Vision
Problem Statement	Classify near-infrared images from a Leap Motion sensor into ten hand gestures for touchless control of devices. Keep at least one subject's images completely out of training and report accuracy on that unseen person.
Classes	10 — Palm, L, Fist, Fist moved, Thumb, Index, OK, Palm moved, C, Down
Dataset	Hand Gesture Recognition Database, LeapGestRecog (Kaggle) — 20,000 images from 10 subjects
Dataset Link	https://www.kaggle.com/datasets/gti-upm/leapgestrecog

SCENE UNDERSTANDING (PS-50 to PS-50)

PS-50 Natural Scene Classification

Domain	Scene Understanding
Problem Statement	Classify natural and urban scenes into six categories for photo organisation and tourism apps. Inspect your misclassified images: some labels are ambiguous (e.g., a glacier in front of mountains), so estimate how much of your error is due to label noise.
Classes	6 — Buildings, Forest, Glacier, Mountain, Sea, Street
Dataset	Intel Image Classification (Kaggle) — about 25,000 images, 150 × 150
Dataset Link	https://www.kaggle.com/datasets/puneet6060/intel-image-classification

Part C — Instructions to Students

These instructions apply to every problem statement in Part B.

1. **Individual task.** Work only on the problem statement allotted to you. Identical notebooks or reports will be treated as malpractice.
2. **Time.** The hands-on work and the two-page report must be completed within two days.
3. **Dataset.** Use the copy of your dataset provided on the lab systems or shared drive. Where the dataset is large, use the balanced subset given by the faculty and state its size in your report.
4. **Tools.** Python with TensorFlow/Keras or PyTorch, in Jupyter Notebook or Google Colab. ImageNet-pretrained models may be used.
5. **Minimum work expected (for every problem statement):**
    a) Load the data; show the number of images per class and a few sample images from each class.
    b) Create training, validation and test splits with a fixed random seed. Follow any splitting rule given in your problem statement (by patient, lesion, subject or driver).
    c) Build a CNN-based classifier — trained from scratch, by transfer learning from a pretrained network, or both.
    d) Train with suitable pre-processing and augmentation, and plot training and validation loss and accuracy.
    e) Evaluate once on the test set: accuracy, per-class precision, recall and F1, macro F1 and the confusion matrix, plus any metric named in your problem statement.
    f) Show at least four misclassified test images and explain the likely reason for each error.
6. **Specific challenge.** Each problem statement names a specific challenge (class imbalance, data leakage, small data, domain shift, shortcut learning, etc.). Address it explicitly; it carries weight in evaluation.
7. **Code repository.** Save the notebook with all outputs, push it to your GitHub repository and put the link in your report.
8. **File names.** Name your files RegisterNo_PSxx (for example, RegisterNo_PS07_Report.pdf and RegisterNo_PS07.ipynb).
9. **Academic integrity.** Cite any external code or tutorial you adapt, and declare any use of AI assistants in the report.

Part D — Two-Page Report Format

Every student submits a report of not more than two pages. The report is evaluated together with the notebook.

Header block (top of page 1)

Name | Register No. | Section | PS No. and Title | Date | GitHub link to the notebook

Sections

Section	What to write	Approx. length
1. Problem and Objective	The problem in your own words, why it matters, and what success means (the metric you will judge by).	3–4 lines
2. Dataset and Pre-processing	Source and link, classes, number of images used (full set or subset), split strategy and split sizes, resizing and normalisation, augmentation, how class imbalance was handled.	5–7 lines
3. Methodology	Architecture (from scratch or pretrained backbone), layers frozen or fine-tuned, loss function, optimiser, learning rate, batch size, epochs, early stopping. Give hyperparameters in a small table.	6–8 lines + table
4. Results	One results table (model, test accuracy, macro F1, key per-class values, training time) and one or two figures (learning curves and/or confusion matrix).	Table + 1–2 figures
5. Analysis and Discussion	Which classes are confused and why, signs of over- or under-fitting, misclassified examples, and how you handled the specific challenge in your problem statement.	6–8 lines
6. Real-world Relevance, Limitations and Ethics	Where the model could be used, what stops it from being deployed today, and any bias, privacy or safety concerns.	3–5 lines
7. Conclusion and Future Work	Your key result in one sentence and one concrete improvement you would try next.	2–3 lines

Formatting rules

- A4 size, maximum two pages including all tables and figures; no separate cover page.
- Times New Roman 11 pt for text, 12 pt bold for section headings, single line spacing, 2 cm margins on all sides.
- Number and caption every table and figure (e.g., Table 1, Fig. 1); axis labels and legends must be readable when printed.
- Report numbers on the test set only, to two decimal places, and say which split each number comes from.
- Cite the dataset (and its paper, where the dataset page asks for it) in one line at the end of the report.
- Submit as PDF, named RegisterNo_PSxx_Report.pdf.

Part E — Evaluation Rubric

The rubric follows the CBL allocation: Problem Understanding & Planning 10, Technical Approach & Implementation 10, Innovation & Use of AI/ML 5, Results, Evaluation & Impact 3, Presentation & Documentation 2.

Criterion (Marks)	Excellent	Good	Satisfactory	Needs Improvement
Problem Understanding & Planning 10 marks	**9–10 marks** Objective and its importance stated clearly; EDA shows class balance and samples; dataset risks (imbalance, leakage, duplicates) identified; suitable metrics chosen with reasons.	**7–8 marks** Clear objective and EDA; main dataset risk identified, but limited justification for the chosen metrics.	**5–6 marks** Objective stated; minimal EDA; dataset risks not discussed.	**0–4 marks** Problem not understood or wrongly framed; no EDA.
Technical Approach & Implementation 10 marks	**9–10 marks** Complete, correct pipeline (split, pre-processing, augmentation, model, training); well-justified architecture or transfer-learning choice; reproducible (fixed seed, versions noted); clean, commented code.	**7–8 marks** Working pipeline with minor issues; reasonable model choice; mostly reproducible.	**5–6 marks** Model trains but with design flaws (e.g., no validation set, augmentation applied to test data, split rule ignored).	**0–4 marks** Code does not run, or results cannot be reproduced from the notebook.
Innovation & Use of AI/ML Techniques 5 marks	**5 marks** Goes beyond a single baseline with a justified idea (e.g., backbone comparison, imbalance handling, Grad-CAM, robustness test, lightweight model) and explains its effect with evidence.	**4 marks** One improvement attempted, with partial analysis of its effect.	**3 marks** Standard approach only, applied correctly.	**0–2 marks** No evidence of independent thinking.
Results, Evaluation & Impact 3 marks	**3 marks** Correct metrics on a held-out test set (per-class, macro F1, confusion matrix, plus any metric named in the PS); honest error analysis; real-world use and limits discussed.	**2 marks** Correct metrics with limited interpretation.	**1 marks** Accuracy only, or validation results reported as test results.	**0 marks** No valid results.

Criterion (Marks)	Excellent	Good	Satisfactory	Needs Improvement
Presentation & Documentation 2 marks	**2 marks** Report within two pages in the prescribed format; labelled, readable figures; notebook and repository clean and complete.	**1.5 marks** Minor lapses in format or figure quality.	**1 marks** Exceeds the page limit or figures unreadable.	**0–0.5 marks** Report missing or largely incomplete.
TOTAL	**30 marks (10 + 10 + 5 + 3 + 2, matching the CBL weightage of 30% of the final grade)**			

Critique focus for evaluators

- **Depth over breadth:** look for conceptual understanding and application, not just tool usage.
- **Evidence-based:** every claim in the report must be backed by a number, plot or example from the notebook.
- **Reproducibility:** the notebook should re-create the reported numbers (fixed seed, saved split, versions noted).
- **Ethics and responsibility:** bias, privacy, safety and the impact of wrong predictions in the real setting.
- **Individual work:** the student should be able to explain every cell of the notebook during the viva.

Part F — Sample Viva / Critique Questions

Evaluators may pick two or three questions per student.

1. Why did you choose this architecture and input size? What would change with a smaller or larger model?
2. How did you split the data, and how did you make sure no test information leaked into training?
3. Which metric best reflects success for your problem, and why is plain accuracy not enough?
4. Which two classes does your model confuse most, and what visual reason explains it?
5. Compare your training and validation curves. Is the model over-fitting? What did you do about it?
6. How did you handle the specific challenge named in your problem statement? What evidence shows it worked?
7. What would you need before this model could be used in the real setting described in your problem statement?
8. Show where in your notebook the result is made reproducible (seed, saved split, versions).

Part G — Notes for the Faculty / Lab Coordinator

- **Allotment:** allot one problem statement per student (for example, in register-number order). For a batch larger than 50, a problem statement may be repeated only in a different lab batch.
- **Pre-stage the datasets:** download every dataset to the lab server or shared drive before the session, along with the pretrained model weights (Keras and PyTorch download them at run time, which lab networks often block).
- **Accounts and licences:** Kaggle datasets need a free Kaggle login; competition data (PS-09, PS-25, PS-41, PS-48) also needs the competition rules to be accepted. TensorFlow Datasets entries (PS-06, PS-07, PS-17) can be cached in advance with tensorflow_datasets. Some sources (e.g., BreakHis, HP Labs Tamil) are for research use only and ask users to cite their paper.
- **Large datasets:** for problem statements with large datasets (e.g., PS-11, 15, 16, 17, 20, 23, 25, 33, 36, 37, 43, 44, 48, 50), prepare a fixed balanced subset (about 300–500 images per class) so that training fits in the session and results are comparable across students.
- **Split rules:** PS-15 (lesion ID), PS-16 (patient ID), PS-48 (driver list) and PS-49 (subject) require grouped splits; check that students follow them.
- **Link check:** dataset links were compiled in October 2026. Open each link once before the session, since dataset pages occasionally move or change version.